

The π -signature is a DERIVED discriminator (F157 = K923 = Cal's fit-guard, one theorem); it resolves count-vs-measure mechanism; muon S1 structurally cleared, exact-argument check pending

Keeper

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K968 — The π -signature is a theorem, not a heuristic

The reconnection (three things are one)

Lyra's read of the Γ_Ω signature closed a loop: - F157 (Lyra): the domain's Gindikin gamma Γ_Ω is π -ful $\frac{2}{3}$ a = $N_c = 3$ is odd. Forced, with a proof — not a rule of thumb. - K923 (Derived): lepton-vs-quark π -parity — leptons carry π , quarks don't. - Cal §117 fit-guard: “carries π = geometric (analytic Γ_Ω); no π = counted ($|S_4|$, Weyl order).”

These are the same theorem. Cal invented nothing ad hoc to catch the muon; he re-derived, as a fit-guard, the π -parity Lyra had already proved. That is the strongest kind of agreement — a skeptic's demotion tool and a derived structure turning out to be one object.

What the discriminator actually discriminates (scope — do NOT over-apply)

The π -signature separates mechanism type, not tier: - π -ful $\frac{2}{3}$ analytic-measure mechanism (the Γ_Ω / Bergman measure of the domain). - π -less $\frac{1}{3}$ combinatorial-count mechanism (a symmetry/capacity count).

Both can be DERIVED. A forced count ($\alpha^{-1} = 137 = N_c^3 \cdot n_C + \text{rank}$, the charge-capacity count — π -less, correctly, and Derived) and a forced measure ($p/e = 6\pi^5$, π -ful, Derived) are both Derived by different mechanisms. So π -less does NOT mean not-Derived. The tool's real use is narrow and sharp:

When a value is expressible BOTH as a count AND as a measure (an ambiguity), the π -content resolves WHICH mechanism is operating.

The muon is the exact case: $24 = |S_4|$ (count, no π) OR $\Gamma(5)$ (measure, π -ful). The muon form $(24/\pi^2)^6$ pairs 24 with $\pi^2 \rightarrow$ the operating mechanism is the measure (Γ_Ω), not the count. The π resolves the count-vs-measure ambiguity that made 24 look like a coincidence.

Audit against K967 (the muon's blind criteria)

- ★ S1 (co-emergence — 24 and π^2 from the SAME Γ_Ω) — STRUCTURALLY CLEARED. By F157/K923, a π -ful form is Γ_Ω -derived; the muon's 24-paired-with- π^2 is therefore the analytic Γ_Ω value, not a count. This firmly seats the muon at IDENTIFIED, off the FITTED floor — the π is a *derived* certificate, not a noticed pattern.
 - Remaining S1 piece (Grace): the EXACT-ARGUMENT check — is the muon's specific $(24/\pi^2)$ literally the Γ_Ω ratio at the muon's *own* address, or π -ful “in general”? Structural π -fulness is necessary; exact-argument identity is the rest of S1.

- S2 (address target-innocent), S3 (exponent $6 = n_C + 1$ derived), S4 (sector consistency $e/\mu/\tau$), S5 (new forced mechanism) — STILL OPEN. These require the discrete-series address, which waits on Cal's reduction level.

Ruling: the muon stays IDENTIFIED / Insight, now with S1 structurally cleared and the bar sharpened to {exact-argument, address, exponent, sector}. This is genuine forward progress along the pre-registered path — not a promotion. Cal #27 guard holds: the π -signature certifies the *mechanism type*; it does not force the *address*, and Derived needs the address.

The generalization (research the team can use now)

The π -signature is a reusable tier tool for the whole scorecard: for any form whose value has a count-vs-measure ambiguity, read its π -content to pin the mechanism. Quick sweep (consistency check, not new rulings): - $p/e = 6\pi^5$ (π -ful $\rightarrow$ measure) — Derived π - muon $(24/\pi^2)^6$ (π -ful $\rightarrow$ measure, ambiguity resolved) — Identified, path π - $\alpha^{-1} = 137$ (π -less $\rightarrow$ count, capacity K701) — Derived-as-count π (correctly π -less) - $\lambda_H = 1/\text{rank}^3 = 1/8$ (π -less $\rightarrow$ count) — the π -tool says “count mechanism”; check its forcing as a count, not a measure. - the mixing angles (rational, π -less $\rightarrow$ count/overlap) — consistent with the Born-overlap count reading.

No re-tiering falls out of the sweep; it CONFIRMS the existing tiers and gives each a mechanism-type label. That label is itself useful for the Ceiling/Value work (K964): a π -ful ceiling is a measure-constraint, a π -less one a count-constraint.

— K968, Keeper, 2026-07-28. The π -signature banked as a derived, scoped discriminator (F157 = K923 = Cal's fit-guard). Muon S1 structurally cleared; stays Identified. See [[Keeper_K966_muon_does_NOT_bank_Derived_reverts_to_Identified_07-28]], [[Keeper_K967_BLIND_pre_registered_kill_survive_criteria_for_the_muon_Gamma_Omega_promotion_committed_by_07-28]].